

Apply filters to SQL queries

Project description

In this activity, I used SQL filters to investigate potential security issues involving login attempts and employee machines. I queried the `log_in_attempts` and `employees` tables using `AND`, `OR`, and `NOT`, as well as `LIKE` for pattern matching. These queries helped identify suspicious login activity and specific groups of employees whose machines may require security updates

Retrieve after hours failed login attempts

```
SELECT *  
FROM log_in_attempts  
WHERE login_time > '18:00'  
AND success = 0;
```

Explanation:

This query retrieves login attempts made after **18:00 (6:00 PM)** where the login was unsuccessful. The `AND` operator ensures that **both conditions** must be true: the attempt occurred after 18:00 and `success` is `0`, meaning the login failed.

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_time > '18:00' AND success = FALSE;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0

Retrieve login attempts on specific dates

A suspicious event occurred on 2022-05-09. Any login activity that happened on 2022-05-09 or on the day before needs to be investigated.

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	0
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0

```
SELECT *
FROM log_in_attempts
WHERE login_date = '2022-05-09'
OR login_date = '2022-05-08';
```

Explanation:

This query retrieves all login attempts from **May 9, 2022, or May 8, 2022**. The **OR** operator is used because a login attempt only needs to match **one of the two dates**.

Retrieve login attempts outside of Mexico

After investigating the organization's data on login attempts, I believe there is an issue with the login attempts that occurred outside of Mexico. These login attempts should be investigated.

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	0
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	0

```
SELECT *
FROM log_in_attempts
WHERE NOT country LIKE 'MEX%';
```

Explanation:

This query retrieves login attempts that did **not** originate from Mexico. **LIKE 'MEX%'** matches values beginning with **MEX**, including both **MEX** and **MEXICO**. The **NOT** operator reverses the condition, so those records are excluded.

Retrieve employees in Marketing

My team wants to update the computers for certain employees in the Marketing department. To do this, I have to get information on which employee machines to update.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Marketing' AND office LIKE 'East%';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267

```
SELECT *  
FROM employees  
WHERE department LIKE 'Marketing%'  
AND office LIKE 'East%';
```

Explanation:

This query identifies employees in the **Marketing department** who work in an office in the **East building**. **LIKE 'Marketing%'** matches department values beginning with **Marketing**, while **LIKE 'East%'** matches office values beginning with **East**. The **AND** operator requires both conditions to be true.

Retrieve employees in Finance or Sales

The machines for employees in the Finance and Sales departments also need to be updated. Since a different security update is needed, I have to get information on employees only from these two departments.

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170

```
SELECT *
FROM employees
WHERE department = 'Finance'
OR department = 'Sales';
```

Explanation:

This query identifies employees who work in either the **Finance** or **Sales** department. The **OR** operator allows either condition to be true.

Retrieve all employees not in IT

My team needs to make one more security update on employees who are not in the Information Technology department. To make the update, I first have to get information on these employees.

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434

```
SELECT *
FROM employees
WHERE NOT department = 'Information Technology';
```

Explanation:

This query retrieves employees who are **not** in the Information Technology department. The

NOT operator reverses the condition and excludes employees whose department is **Information Technology**.

Summary

In this activity, I used SQL filtering techniques to investigate login attempts and employee information related to potential security incidents. I used **AND** and **OR** to combine multiple conditions, **NOT** to exclude specific records, and **LIKE** with **%** to search for patterns in country, department, and office values. These filtering techniques allowed me to identify failed after-hours login attempts, investigate activity on specific dates and outside Mexico, and identify employees whose machines may require security updates